Finite-Attempt Rényi's Parking Problem with various distributions

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Background

The classical **Rényi parking problem** studies the random placement of non-overlapping intervals, or "cars," on a line. Cars are placed sequentially at random positions. If a proposed car overlaps a previously parked car, the placement is rejected.
In the classical model, this process continues until no more cars can fit. The system approaches a limiting density known as the **Rényi parking constant**:

$$\rho \approx 0.7476.$$

This limiting state is called a **jamming state**.

Motivation

The original Rényi model assumes that placement attempts can continue indefinitely.
This project studies a more restricted version:

- Each car is allowed only k placement attempts.
- If all k attempts fail, the car is discarded.
- Different spatial distributions are used to choose attempted positions.

This creates a transition between a **resource-limited** model and a **geometry-limited** jamming model.

Research Questions

- How does the number of attempts k affect final density?
- How do different spatial distributions affect packing?
- Does the finite-attempt model approach the classical Rényi limit?
- What does the gap structure reveal about jamming?

Model

A line segment of length L is filled with cars of length 1. The density is computed as

$$\rho = \frac{\text{total occupied length}}{\text{total line length}}.$$

The attempted starting positions are sampled from:

- Uniform distribution
- Normal distribution
- Exponential distribution
- Gaussian integer mixture

Methodology

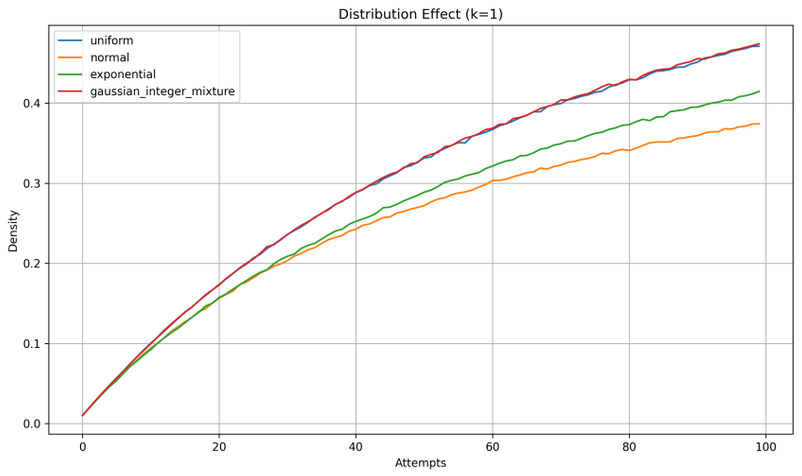
The simulation uses a Monte Carlo approach.
For each distribution and value of k:

1. Start with an empty line segment.
2. Attempt to place cars sequentially.
3. Reject placements that overlap existing cars.
4. Allow each car at most k attempts.
5. Repeat the experiment many times.
6. Average the final density and gap statistics.

The key parameters were:

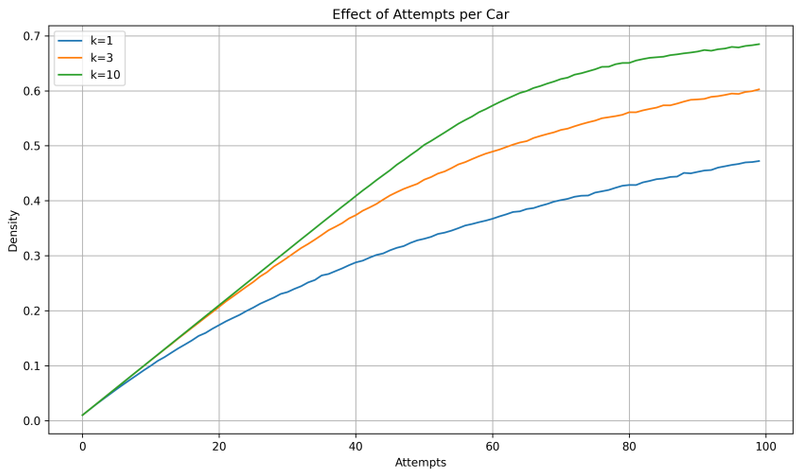
Parameter	Value
Line length	100
Car length	1
Attempts k	$1, 2, 3, 5, 10, 20$
Trials per setting	300 to 1000

Distribution Effect When $k = 1$



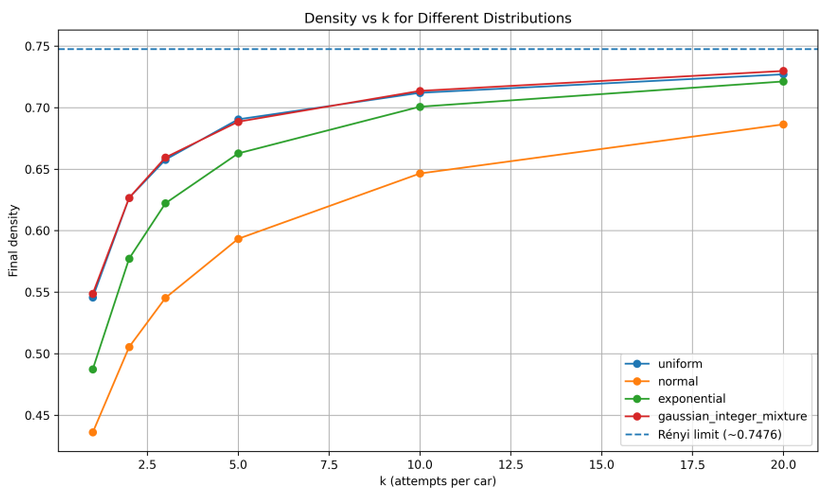
When $k = 1$, each car has only one chance to park. This makes the system highly sensitive to the distribution of attempted positions.
Uniform and Gaussian mixture sampling perform best, while normal sampling performs worst because it creates congestion near the center of the interval.

Effect of Attempts per Car (Uniform)



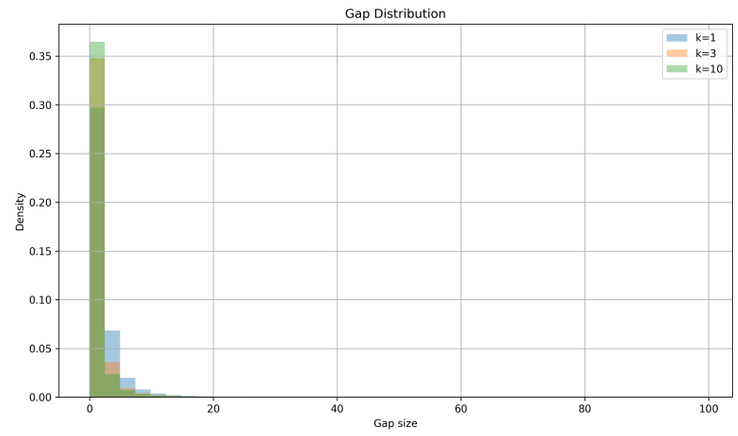
Increasing k greatly improves packing efficiency. Higher values of k allow rejected cars to search again, reducing wasted space and increasing final density.

Density vs. k Across Distributions



All distributions show increasing final density as k increases.
At low k, distribution choice matters strongly. At high k, the curves move closer together and approach the Rényi limit.

Gap Distribution



As k increases, the system produces more small gaps and fewer large gaps.
This is evidence of **jamming**: empty spaces remain, but many are too small to fit another car.

Conclusion/Key Findings

- Density increases monotonically with k.
- Distribution choice matters most when k is small.
- Larger k values reduce the effect of spatial bias.
- Gap distributions shift toward smaller intervals as jamming emerges.
- The finite-attempt model trends toward the Rényi density.

This project extends the classical Rényi parking problem by limiting the number of placement attempts per car.
The results suggest two regimes:

Regime	Dominant Factor
Low k	Spatial distribution
High k	Geometry / jamming

Thus, increasing k shifts the system from a **resource-limited** process to a **geometry-limited** jamming process.

Future Work

- Study larger intervals to reduce finite-size effects.
- Analyze convergence rate toward the Rényi constant.
- Extend the model to two-dimensional parking.
- Compare simulation results with analytic approximations.

References

[1] A. Slavík, *De Bruijn's Short Route to Rényi's Parking Constant*, 2024.

[2] J. Kiely, *The Parking Problem: Part 3 – Rényi's Approach*, 2023.

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